

Emanuele Bugliarello

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Research Interests

Multimodal and multilingual natural language understanding, representation learning, transfer learning

Education

PhD in Computer Science 2019–present
University of Copenhagen Copenhagen, Denmark
Advisors: Desmond Elliott, Anders Søgaard

MSc in Communication Systems, Data Analytics specialization 2015–2018
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Thesis: *Matrix Completion in the Unit Hypercube via Structured Matrix Factorization*
GPA: 5.53/6 (3/44)

BEng in Electronics and Information Engineering 2013–2018
Tongji University (同济大学) Shanghai, China
Thesis: *General large-batch methods for scalable accurate training of neural networks*
GPA: 29.34/30

BSc in Telecommunications Engineering 2012–2015
Politecnico di Torino Turin, Italy
GPA: 28.86/30 (3/120) – Graduation Grade: 110/110 *Cum Laude*

Machine Learning Schools: Lisbon Machine Learning School (LxMLS) 2020, Oxford Machine Learning Summer School (OxML) 2020, Advanced Language Processing Winter School (ALPS) 2021

Experience

Research Intern with Rishabh Mehrotra and Mounia Lalmas 08/2020–11/2020
Spotify Research London, UK
On Multi-Objective Music Sequencing

- Propose counterfactual beam search to boost platform objectives while controlling for customer satisfaction

Research Associate with Prof. Naoaki Okazaki 10/2018–09/2019
Okazaki Lab @ Tokyo Institute of Technology (東京工業大学) Tokyo, Japan
On Self-Attentive Neural Machine Translation

- Design of a dependency-aware self-attention and cross-lingual evaluation via cross-mutual information

Visiting Scholar with Prof. Rui Fan and Prof. Dan Alistarh 04/2018–07/2018
ShanghaiTech University (上海科技大学) Shanghai, China
Large-Batch Algorithms for Deep Learning at Scale

- Implementation and generality evaluation of state-of-the-art algorithms for large-batch training of CNNs

Research Intern with Swayambhoo Jain, Brian Eriksson and Jean Bolot 09/2017–03/2018
Technicolor AI Lab Palo Alto, CA, USA
Intelligent VFX Scheduling

- Propose constrained matrix factorization models to recommend artists to visual effects (VFX) departments

Research Assistant with Marcel Schöngens, Maxime Martinasso and Claudio Gheller Summer 2017
Swiss National Supercomputing Centre (CSCS) Lugano, Switzerland
Distributed TensorFlow: A performance evaluation

- Scalability study of distributed TensorFlow using InceptionV3 for image classification on ImageNet

Teaching

Teaching Assistant, EPFL: Internet Analytics (COM-308) Spring 2017

Teaching Assistant, UCPH: Advanced Topics in Natural Language Processing (NDAK19001U) Winter 2020

Teaching Assistant, UCPH: Neural Information Retrieval (NDAK20002U)

Spring 2021

Publications

5. **Multimodal Pretraining Unmasked: A Meta-Analysis and a Unified Framework of Vision-and-Language BERTs**

E. Bugliarello, R. Cotterell, N. Okazaki, D. Elliott
TACL 2021

4. **The Role of Syntactic Planning in Compositional Image Captioning**

E. Bugliarello, D. Elliott
In EACL 2021

3. **It's Easier to Translate *out of* English than *into* it: Measuring Neural Translation Difficulty by Cross-Mutual Information**

E. Bugliarello, S.J. Mielke, A. Anastasopoulos, R. Cotterell, N. Okazaki
In ACL 2020

2. **Enhancing Machine Translation with Dependency-Aware Self-Attention**

E. Bugliarello, N. Okazaki
In ACL 2020

1. **Matrix Completion in the Unit Hypercube via Structured Matrix Factorization**

E. Bugliarello, S. Jain, V. Rakesh
In IJCAI 2019

Skills

Programming

- Python, C, Java, Matlab, Java for Android
- NumPy, Pandas, scikit-learn, CUDA
- PyTorch, TensorFlow
- Spark, Hadoop MapReduce, MPI

Languages

- native: Italian
- fluent: English (TOEFL iBT 109)
- intermediate: French
- basic: Chinese (HSK III), Japanese, Danish

Fellowships, Honors, and Awards

EuroHPC Grant with Desmond Elliott on “Vision and Language on a Shoestring without Tear”	06/2021
University of Copenhagen Marie Skłodowska-Curie TALENT Doctoral Fellowship	06/2019
EPFL PhD Program in Computer Science Fellowship [offer declined]	05/2019
EPFL TCP/IP 2015-2016 Research Exercise Award	12/2015
PoliTong Merit Scholarship	07/2013
MIUR National Register of Excellences	06/2012

Service

Reviewer: EMNLP, ICLR, NeurIPS